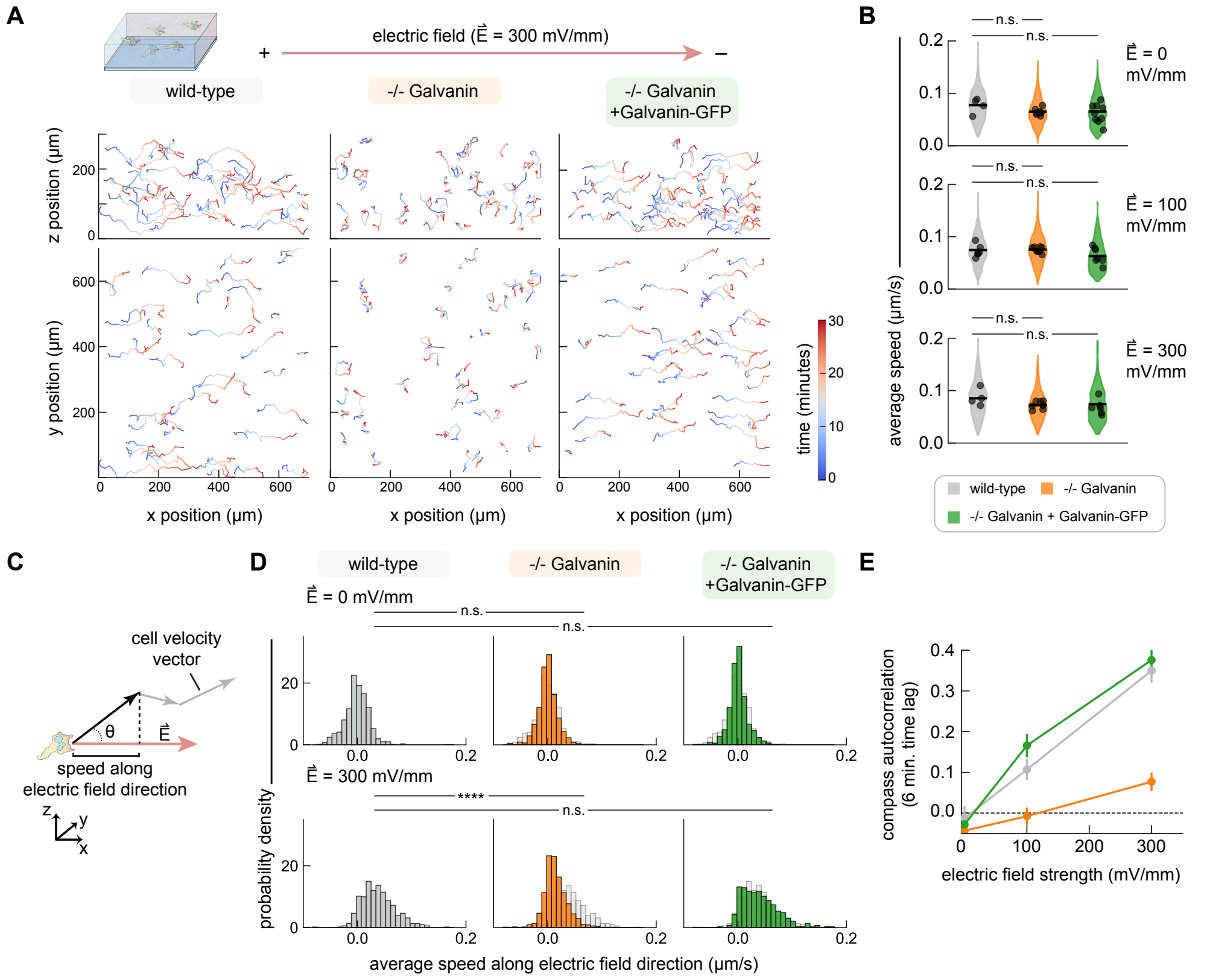

A
electric field (E⃗ = 300 mV/mm)
+
−
wild-type
-/- Galvanin
-/- Galvanin
+Galvanin-GFP
z position (μm)
y position (μm)
x position (μm)
time (minutes)
B
n.s.
E⃗ = 0 mV/mm
E⃗ = 100 mV/mm
E⃗ = 300 mV/mm
average speed (μm/s)
wild-type
-/- Galvanin
-/- Galvanin + Galvanin-GFP
C
cell velocity vector
θ
E⃗
speed along electric field direction
D
probability density
average speed along electric field direction (μm/s)

E
compass autocorrelation (6 min. time lag)
electric field strength (mV/mm)